

Circuit Analysis (Lab11)

Experiment 10: Verification of Norton's Theorems

Name: Registration No:

Date: Grade and Signature:

CLO2 CLO3	Use with identification the circuit components, breadboards, multimeter, power supplies, signal generators, and oscilloscopes. Make measurements in their own constructed electric circuits.			
Psychomotor/Affective	Level1 (1)	Level2 (2-3)	Level3 (4-5)	Level4 (6-7)
Report Marks (3)				
	Total marks (10)			

Objective

The objective of this lab is verify Norton's theorem and to find the full load current.

Apparatus Required

Sr. no	Apparatus	Range	Quantity
1	Regulated Power Supply	Standard	1
2	Signal Generator	Standard	1
3	Oscilloscope	Standard	1
4	Digital Multimeter	Standard	1
5	Breadboard & Wires	Standard	As Required
6	Resistors	1k, 330 ohms	1,3
7	Potentiometer	Standard	1

Description

Statement:

Any linear circuit can be simplified to an equivalent circuit consisting of a single current source and parallel resistance that is connected to a load.

Precautions:

1. Voltage control knob of RPS should be kept at minimum position.
2. Current control knob of RPS should be kept at maximum position

Procedure for Norton's Theorem:

1. Connections are given as per circuit diagram.
2. Set a particular value in RPS and note down the ammeter readings in the original circuit.

To Find I_N :

3. Remove the load resistance and short circuit the terminals.
4. For the same RPS voltage note down the ammeter readings.

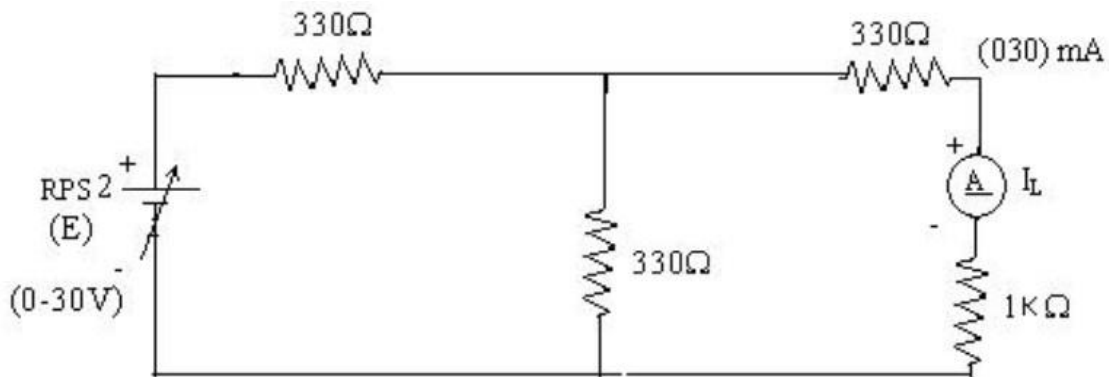
To Find R_N :

5. Remove RPS and short circuit the terminal and remove the load and note down the resistance across the two terminals.

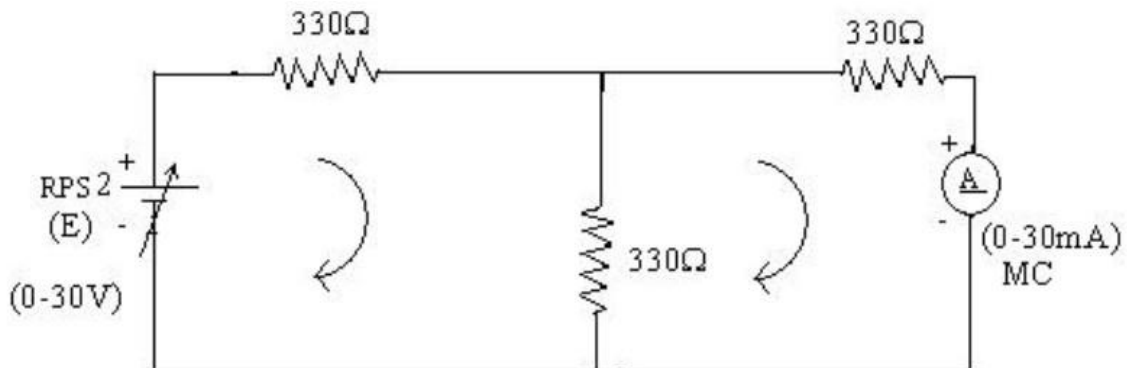
Equivalent Circuit:

6. Set I_N and R_N and note down the ammeter readings.
7. Verify Norton's theorem.

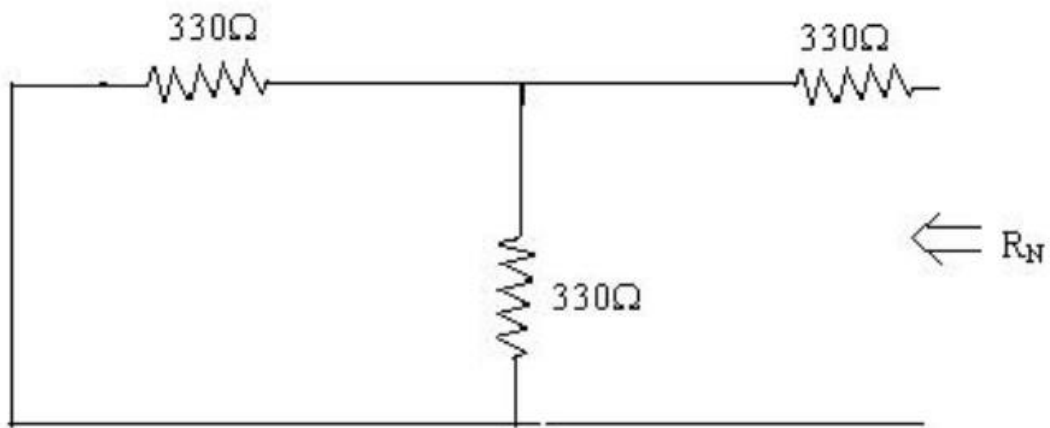
To find load current in circuit 1:



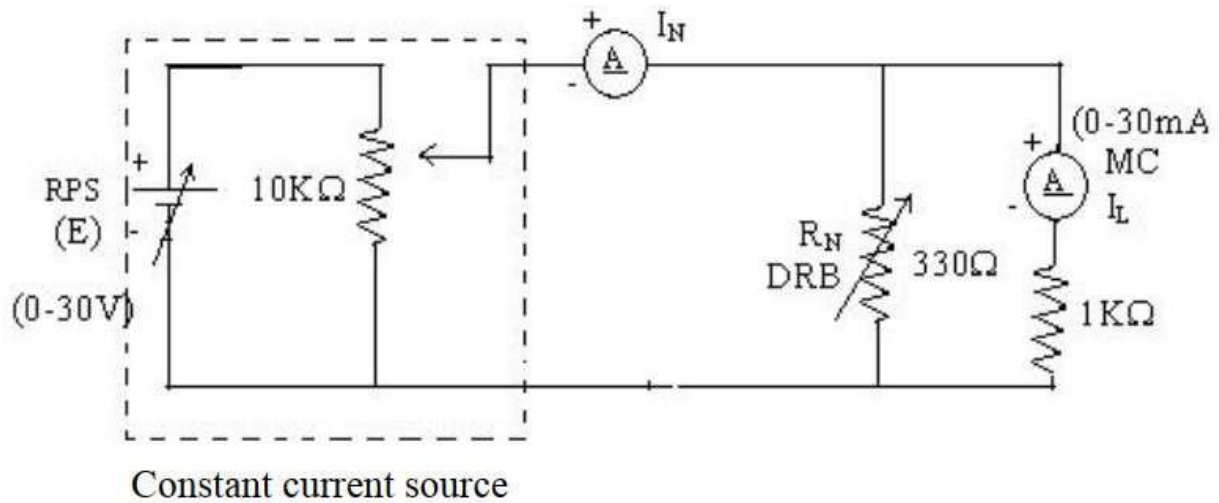
To find I_N



To find R_N



Norton's equivalent circuit



Norton's Practical Values

	E (volts)	I _N (mA)	R _N (ohms)	I _L (mA)	
				Circuit - I	Equivalent Circuit
Practical Values	10 V	10.10	495	3.34	3.34

Post Lab Questions

1. What is Norton's Theorem?

Norton Theorem: Any linear bilateral circuit can be replaced by an equivalent circuit

consisting of a single constant current source I_N in parallel with resistance R_N .

I_N = short-circuit current at load terminals (load removed, terminals shorted).

R_N = equivalent resistance seen from load terminals with all sources deactivated

(voltage sources shorted, current sources open-circuited).

Load current $I_L = I_N \times R_N / (R_N + R_L)$. Dual of Thevenin's Theorem: $R_N = R_{TH}$.

Simplifies complex networks to a 2-element equivalent for any load analysis.

Verified: $I_L(\text{Circuit-I}) = I_L(\text{Equivalent}) = 3.34 \text{ mA}$ at $E = 10\text{V}$.

2. What are the applications of Norton's Theorem?

1. Simplifying power distribution networks to find load current easily.

2. Analyzing transistor amplifier circuits and determining bias currents.

3. Maximum power transfer calculations in communication and RF systems.

4. Short-circuit fault current analysis in electrical power systems.

5. Source impedance matching in signal processing and filter design.

6. Reducing large sub-networks in SPICE/circuit simulation software.

7. Sensor circuit analysis to determine equivalent current output.

8. Used in network analysis for two-port equivalent circuit modeling.

3. What are the theoretical and practical limitations of Norton's Theorem?

THEORETICAL: 1. Only valid for linear bilateral circuits · fails for diodes,

transistors (non-linear), or magnetic saturation elements.

2. Requires careful handling when dependent sources are present in the network.

3. Assumes DC steady-state conditions; not valid for transient analysis.

4. Not directly applicable to time-varying or frequency-dependent elements.

PRACTICAL: 5. Resistor tolerances ($\pm 5\text{-}10\%$) cause I_N and R_N to deviate from theory.

6. Breadboard contact resistance and lead resistance introduce measurement error.

7. Ammeter internal resistance affects measured I_N and I_L current values.

8. RPS internal resistance causes voltage droop under different load conditions.

9. Temperature coefficient of resistors changes R_N value at higher power levels.

10. Stray capacitance and inductance affect results at higher frequencies.

Conclusion: $I_L(\text{Circuit-I}) = I_L(\text{Equivalent}) = 3.34 \text{ mA}$. Theorem verified.